Md Zahidul Islam

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Professional Summary

- Ph.D. candidate in Industrial Engineering (Operations Research) with 9+ years of experience building data-driven models that support decisions in complex, large-scale operations.
- Builds analyses in Python that combine machine learning and statistical modeling with optimization (mixed-integer and multi-objective programming in Gurobi), Markov decision processes, and reinforcement learning, for workforce scheduling, healthcare operations, demand forecasting, supply chain, and routing.
- Published researcher who works with industrial engineers, industrial-organizational psychologists, emergency physicians, and sleep scientists and communicates findings to technical and non-technical audiences.

Technical Skills

Programming:	Python, SQL, R, MATLAB, Git, SLURM/HPC
Python Libraries:	pandas, NumPy, SciPy, scikit-learn, XGBoost, PyTorch, TensorFlow, NetworkX
Optimization Modeling:	Linear, Mixed-Integer, Constraint, and Multi-Objective Programming; Markov Decision Processes and Dynamic Programming; Network Optimization; Queueing; Gurobi, CPLEX
Data Science:	Regression and Classification, Gradient Boosting, Time-Series Forecasting, Deep Learning (Neural Networks, LSTM), Reinforcement Learning, Model Tuning and Validation
Simulation & Statistics:	Agent-Based and Discrete-Event Simulation (AnyLogic, Arena); Statistical Inference, Survey Data Analysis, Design of Experiments; SPSS, Minitab, Microsoft Excel

Work & Research Experience

Ph.D. Student Aug 2023 – Present

Department of Industrial Engineering, Clemson University

Emergency Department Admission Prediction *(Spring 2025)*

- Built an end-to-end Python pipeline that joins, cleans, and engineers features from MIMIC-IV-ED tables, turning 425,011 encounters into a 347,360-visit dataset with 50 structured predictors and 500 TF-IDF text features.
- Compared five classifiers tuned by cross-validated grid search; CatBoost reached the best ROC-AUC (0.863), LightGBM gave the highest recall for admitted patients (0.779), and acuity score and age were the strongest predictors.

Multimodal Clinician Data Collection and Analytics *(Jan 2025 – Present)*

- Designed and deployed a multi-site clinician survey, conducted interviews and analyzed them using thematic analysis, and collected hospital operational and EHR data, salivary cortisol, and sleep and fatigue-risk data from the ReadiWatch wearable to study how shift schedules affect clinician fatigue and burnout.
- Cleaned and linked these sources into analysis-ready datasets, built an objective schedule–preference alignment score, and analyzed the data with statistical models, finding that preferences vary widely within each profession and that better-aligned schedules were associated with lower burnout and less end-of-shift fatigue.
- Translated the findings into scheduling-model requirements (protective rules for all clinicians, profession-specific rules, and individual preferences) and presented them to health-system leadership.

Multi-Hospital Physician Scheduling with Heterogeneous Preferences *(Feb 2026 – Present)*

- Developed a MILP model in Gurobi that builds one monthly schedule for emergency physicians rotating across multiple ED facilities, giving each physician a fixed budget of preference points to spread across the scheduling characteristics they value most, such as shift type, weekends, work–rest patterns, and site.
- Enforced coverage, rest, and contract-hour rules, penalized uneven night and weekend loads among peers, and modeled work–rest patterns with network flow; on generated test months, the characteristics each physician weighted most were honored most often in their schedule, all dated requests were granted, and nights were shared more evenly.

Fatigue-Aware Multi-Objective Rostering *(Mar 2025 – Present)*

- Formulated a multi-objective rostering model with a chronotype-sensitive shift fatigue index, balancing fatigue, preferences, and fairness over a 90-day horizon; in a generated test instance, prioritizing fatigue cut the modeled index by about 46% but shortened preferred work blocks, a tension clinicians had described in interviews.

Dynamic Physician Allocation using Markov Decision Processes *(Fall 2025)*

- Solved a queueing-based discounted MDP (up to 5,041 states) by value, policy, and modified policy iteration and a Gurobi linear program; the optimal policy shifted physicians to critical care as that queue grew.

M.S. Student Aug 2021 – Aug 2023

Department of Industrial and Systems Engineering, University of Tennessee, Knoxville

Bike-Sharing Demand Forecasting *(Spring 2023)*

- Forecast hourly bike-rental demand from weather, time, and seasonal features with regression, neural network, LSTM, XGBoost, and random forest models; the tuned neural network explained 96% of test variance and cut linear-regression error by more than half; hour, temperature, season, and humidity were the main demand drivers.

Supervised Learning for the Traveling Salesman Problem *(Aug 2022 – Aug 2023)*

- Collaborated on a supervised-learning method that uses local distance features to predict parts of high-quality tours and guide exact and heuristic solvers, reducing the optimality gap by 28% and runtime by 59% versus a graph pointer network on large instances (IEEE Transactions on Intelligent Transportation Systems).

Agent-Based Supply Chain Simulation for the Bullwhip Effect *(Aug 2021 – Aug 2022)*

- Built an agent-based simulation of a multi-stage supply chain in AnyLogic to test ways to dampen the bullwhip effect, where small changes in customer demand grow into larger order swings upstream; combining information sharing and risk pooling cut order variability by 46–49% at the factory and wholesaler (Winter Simulation Conference).

Deep Reinforcement Learning for Perishable Inventory Optimization *(Jan 2022 – Dec 2022)*

- Modeled perishable-inventory ordering under stochastic demand, deterioration, and quantity discounts as a Markov decision process and trained DQN, A2C, and PPO agents to learn ordering policies for FIFO and LIFO issuing.

Education

Ph.D. in Industrial Engineering (Operations Research) Aug 2023 – Expected Dec 2027

Clemson University | GPA: 3.90/4.00

M.S. in Industrial Engineering Aug 2021 – Aug 2023

University of Tennessee, Knoxville | GPA: 4.00/4.00

Relevant Coursework: Integer Programming, Nonlinear Programming, Stochastic Processes, Markov Decision Processes, Reinforcement Learning, Advanced Optimization via Simulation, Applied Data Science, Advanced Engineering Economy

Leadership Experience

President, INFORMS Student Chapter, Clemson University Aug 2025 – Sep 2026

- Organized invited seminars, professional-development, and outreach events; co-hosted the *Optimality Gap* podcast. The chapter received the INFORMS Magna Cum Laude award in 2025 (as Vice President) and 2026 (as President).

Undergraduate Research Mentor, Clemson Creative Inquiry Fall 2025 – Present

- Mentor twelve undergraduates from industrial engineering, biological sciences, bioengineering, health science, and nursing in field data collection, data-quality checks, analysis, and research presentation.

Conference Presentations & Posters

- Emergency department physician scheduling with heterogeneous individual preferences, 2026 INFORMS Annual Meeting (accepted oral presentation)
- Preferences, alignment, fatigue, and burnout: a cross-sectional shift-schedule survey of emergency department nurses and attending physicians, Prisma Health Research Showcase 2026 (poster)
- Multi-site physician scheduling incorporating work-related fatigue and individual preferences, 2026 Institute of Industrial and Systems Engineers Annual Conference & Expo (oral presentation)
- Fatigue-aware multi-site rostering model for emergency department physicians with fairness and well-being constraints, Healthcare Systems Process Improvement Conference 2026 (oral presentation)
- The impact of emergency department shift scheduling on clinician fatigue and well-being: a qualitative study, Prisma Health Research Showcase 2025 (poster)

Selected Publications

- Lyu, Z., Islam, M. Z., & Yu, A. J. (2024). A scalable and adaptable supervised learning approach for solving the traveling salesman problems. *IEEE Transactions on Intelligent Transportation Systems*, 25(11), 17092–17104.
- Islam, M. Z., Roozeboom, N., Gunderson, P., Li, X., & Yu, A. J. (2022). An agent-based simulation model to mitigate the bullwhip effect via information sharing and risk pooling. *2022 Winter Simulation Conference*, 1636–1647.

Honors & Awards

- University Gold Medal (2019) and Prime Minister Gold Medal (2018), received for the highest CGPA during undergraduate studies at Rajshahi University of Engineering & Technology
- Dr. Lloyd W. Crawford Fellowship, University of Tennessee Space Institute (2021)
- CECAS GSAB Travel Grant and Graduate Student Government Graduate Travel Grant, Clemson University (2026)